TAMONGSCO HIGHLANDS ZONE
ZONAL ACADEMIC COMMITTEE
FORM FOUR MOCK EXAMINATIONS – SEPTEMBER, 2022

CODE: 032 /2A **CHEMISTRY 2A**
(ACTUAL PRACTICAL)
(For both school and private candidates)

MARKING SCHEME

1. (i) Volume of pipette $20cm^3$
NB: burette reading should be equivalent to volume of pippete uses
Table of results

Titration	Pilot	1	2	3
Final reading (mls)	10.00	20.00	30.00	40.00
Initial reading (mls)	0.00	10.00	20.00	30.00
Volume of acid used(mls)	10.00	10.00	10.00	10.00

Table =o1marks

Decimal place = 01marks
Accurate data = 03 marks

Total =05 marks

Average volume of acid used = $\frac{V1+V2+V3}{3}$

$$= \frac{10+10+10}{3}$$

$$= \frac{30}{3}$$

$$= 10cm^3$$

∴ The average volume of acid used is = $10cm^3$ **01 mark**

(ii) Volume of pipette $25cm^3$

Titration	Pilot	1	2	3
Final reading (mls)	12.50	25.00	37.50	12.50
Initial reading (mls)	0.00	12.50	25.00	00.00
Volume of acid used(mls)	12.50	12.50	12.50	12.50

(a) Acid is **GO solution** and base is **EX solution** **@0.5 = 01 mark**

(b) (i) The colour change at the neutralization point was **Yellow colour** to **Orange colour** **@0.5 = 01 marks**

(ii) **$20cm^3$** or **$25cm^3$** of solution **EX** requires **$10cm^3$** or **$12.5cm^3$** of solution **GO** to complete neutralization.
@0.5 = 01 marks

(c) The balanced chemical Equation for the reaction

$$2KOH_{(aq)} + H_2SO_{4(aq)} \longrightarrow K_2SO_{4(aq)} + 2H_2O_{(l)}$$ **02 mark**

Mole ration bases : Acid = 2 : 1

(d) Molarity of base (KOH)

Data Given

Number of particles = 1.505×10^{22} Particles

Volume of solution = $500cm^3$ or $0.5dm^3$

Avogadro's constant = 6.02×10^{23}

Required: Molarity of a base.

But

$$\text{Number of moles (n)} = \frac{Number\ of\ particles(N)}{avogadro's\ number\ (L)}$$

$$= \frac{1.505 \times 10^{22}}{6.02 \times 10^{23}} \quad \textbf{01 mark}$$

$$= 0.025mol \quad \textbf{01 mark}$$

$$\text{Molarity} = \frac{Number\ of\ moles}{volume}$$

$$= \frac{0.025mol}{0.5dm^3} \quad \textbf{01 mark}$$

$$= 0.05M$$

∴ Molarity of a base (KOH) = 0.05M **01 mark**

(d) Calculate the Molarity of sulphuric acid used to react with a base**.**

- Data for finding molarity of acid

  If pipette used was $20cm^3$ then

  $V_a = 10cm^3$

  $V_b = 20\ cm^3$

  $M_b = 0.05M$

  $n_a = 1$

  $n_b = 2$

  $M_a = ?$

$$\text{Formula} = \frac{Ma\ Va}{na} = \frac{Mb\ Vb}{nb}$$

$$Ma = \frac{Mb\ Vbna}{Vanb}$$

$$= \frac{0.05x\ 20x1}{10x2} \quad \textbf{01 mark}$$

$$= \frac{1}{20}$$

$$= 0.05M$$

Therefore, the Molarity of acid(Ma) = $0.05mol/dm^3$ **01 mark**

(d) (i) The concentration of solution changed with time because sulphuric acid is not primary standard Chemical substance. **01 mark**

Or

The concentration of sulphuric acid solution in two months ago it was **decreased** due to absorption of water vapour from the surroundings.

(ii) This acid solution is not recommended for the preparation of standard solution in the laboratory due to the followings reasons

- ✓ It has low degree of purity.
- ✓ ii. It decompose with time
- ✓ it is **hygroscopic** substance that dissolves water to form solution
- ✓ It is volatile so that losses due to evaporation do not occur.
- ✓ Have low molecular mass. **Any 4 @01 = 04mark**

(iv) The concentration (in Mol/dm^3) of the resulting solution?

Data

- Let V_1 and M_1 Stands for Volume and molarity of first solution of H_2SO_4
  Such as, $V_1 = 20cm^3$
  $M_1 = 0.05M$
- Let V_2 and M_2 stand for volume and molarity of second solution of H_2SO_4 Such as,
  $V_2 = 20cm^3$
  $M_2 = 0.8M$
- Let V_c and M_c stand for volume and molarity of the resulting solution after mixing respectively
  Such as, $V_c = V_1 + V_2$
  $= 20cm^3 + 20cm^3$
  $= 40cm^3$
  M_c = ? to be calculated **01 mark**

Formula

Number of moles = Molarity of solution x Volume of solution

$n_1 = MV$

$n_c = n_1 + n_2$

Formula, $McVc = M_1V_1 + M_2V_2$

$$Mc = \frac{M1V1 + M2V2}{Vc}$$

$$= \frac{(0.05\ x\ 20) + (0.8\ x\ 20)}{40}$$ **01 mark** or $= \frac{(0.05\ x\ 0.02) + (0.8\ x\ 0.02)}{0.04}$

=0.425M

There fire, the concentration of the resulting solution = 0.425M mol/dm^3 01 mark

2. (i) Table of results

S/N	Experiment	Observation	Inference
(a)		D was colourless solution	NH_4^+, Zn^{2+}, Ca^{2+}, Na^+ or Pb^{2+} may be present OR Transition elements Fe^{2+}, Fe^{3+}, Cu^{2+} may be absent
(b)		No precipitate formed	NH_4^+, Ca^{2+}, Na^+ may be present
(c)		No precipitate formed on warming a colourless gas evolved which turned moist litmus paper from red to blue	$NH4^+$ confirmed
(d)		A white precipitate was formed	SO_4^{2-} or Cl^- or CO_3^{2-} may be present
(e)		White precipitate formed insoluble in dilute nitric acid	SO_4^{2-} confirmed

Observation 05 @ = 01 marks
Inference 05 @ = 01 marks

Conclusion

- Cation in D is $\underline{NH_4^+}$ **01 mark**
- Anion in D is $\underline{SO_4^{2-}}$ **01 mark**
- Formula of compound D is $\underline{(NH4)_2SO_4}$ **02 marks**
- The chemical name of compound D is Ammonium sulphate **02marks**

(b) (i) $Pb^{2+}_{(aq)} + SO_4^{2-} \longrightarrow PbSO_{4(s)}$ **01 mark**

(ii) **Equation for experiment C**

$2NaOH + (NH4)_2SO_4 \longrightarrow Na_2SO_4 + 2H_2O + 2NH_3$ **02 mark**

(ii) **Equation for experiment e**

$(NH_4)_2SO_4 + Ba(NO_3)_2 \longrightarrow BaSO_{4(s)} + NH_4NO_{3(aq)}$ **02 mark**

(c) Four (4) uses of the reduction of cation in solution **D** in daily life.

- ✓ It is used in manufacturing of fertilizers
- ✓ It is used to softens water and neutralizes acid stains (cleaning)
- ✓ It is used as a refrigerant due to its highly volatile.
- ✓ It is used in manufacturing of nitric acid **Any 4 @01 = 04mark**